

Notes: Berk, van Binsbergen, and Liu (JF, 2017)

Matching Capital and Labor

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1 Big picture

- **Question.** Why do firms exist in an industry where firms own little physical capital and where outside investors can directly observe worker output? The paper answers this by studying mutual fund companies.
- **Core idea.** A key role of the firm is to match capital to labor. In this setting, the firm decides how much assets under management (AUM) each portfolio manager should manage.
- **Why mutual funds are ideal.** The employee's task is clear: generate returns on capital. The amount of capital assigned to the employee is observable. Output can be measured using value added relative to passive Vanguard benchmark funds.
- **Empirical object.** The paper studies internal capital reallocation decisions: promotions are increases in a manager's AUM caused by assigning additional funds; demotions are reductions in a manager's AUM caused by removing funds.
- **Main finding.** Internal firm capital reallocations increase manager value added. The baseline reallocation dummy adds about \$681,000 per manager-month; promotion estimates are around \$0.76–0.88 million per month in the main specifications.
- **Lower-bound contribution.** Under conservative counterfactuals, mutual fund firms add at least \$474,000 per manager-month, more than 30% of the industry's average manager-level value added.
- **Mechanism.** The firm has private information about managerial skill. Fund company executives observe trades, unexecuted ideas, internal process, risk systems, and organizational information that outside investors cannot observe.
- **Investor response.** Investors reward firms after capital reallocations by allocating additional money to the firm's funds. The firm captures value through higher AUM and therefore higher fees.
- **What investors cannot do.** Past performance and other public information explain investor flows, but they explain very little of firms' internal reallocation decisions. External reallocations are not value-adding in the same way as internal reallocations.
- **Important measurement lesson.** Gross alpha is not the right measure of the value created by reallocation. A promotion can reduce gross alpha because of decreasing returns to scale while increasing total value added.
- **Economic takeaway.** Firms can be valuable even without owning capital. Their value can come from privately informed assignment of responsibility: deciding which worker should control how much capital.

2 Incremental contribution of the paper

2.1 Contribution relative to the theory of the firm

Traditional theories of the firm emphasize ownership of assets, incomplete contracts, property rights, or authority over physical capital. This paper studies an industry in which mutual fund companies generally do not own the capital they manage. Investors own the capital and can withdraw it. This makes the mutual fund industry a sharp setting for isolating a non-ownership role of firms.

Incremental contribution: The paper identifies a role of the firm that is distinct from owning capital: the firm creates value by assigning capital to labor based on private information about worker skill. The firm is valuable because it decides the *scope* of each worker's job, not just because it hires the worker.

2.2 Contribution relative to internal labor markets

A large personnel-economics literature studies how firms match workers to jobs, supervisors, and wages. Much less evidence exists on whether firms also use private information to determine how much responsibility a worker should have within a job.

Incremental contribution: The paper provides direct evidence that firms add value by reallocating responsibility within the same occupation. A mutual fund manager’s job is always to manage money, but the firm changes the amount of money the manager manages. This is an unusually clean measure of job scope.

2.3 Contribution relative to mutual fund performance research

Much mutual fund research measures skill using alpha or flow-performance sensitivity. Berk and van Binsbergen’s value-added approach argues that skill should be measured by dollars of value created rather than percentage alpha.

Incremental contribution: This paper applies and extends the value-added approach to the organization of fund families. It shows that capital reallocation can raise total value added even when gross alpha falls. Therefore, measuring only alpha can falsely interpret a successful promotion as a deterioration in skill.

2.4 Contribution relative to internal capital markets

Internal capital markets are usually studied inside diversified industrial firms, where headquarters reallocates capital across divisions or plants. These settings often confound capital ownership, physical assets, and project-level investment opportunities.

Incremental contribution: The mutual fund setting allows the authors to study internal capital allocation without physical capital ownership. The “projects” are managers, and output is directly observed. This gives a cleaner test of whether firms improve allocation.

2.5 Contribution relative to delegated investment and investor attention

The paper also contributes to the delegated investment literature by showing that investors respond to the firm’s internal decisions. Investors appear to treat a reallocation as a signal that the firm has valuable private information.

Incremental contribution: The firm not only creates value internally but also captures part of the value externally. Investor flows into the firm after reallocations show that the market rewards the firm’s private information and organizational decisions.

3 Data and measurement

3.1 The unit of analysis

The paper works with mutual fund firms, funds, and managers. A fund belongs to a firm. A manager may manage one or more funds, and a fund may be comanaged by multiple managers. The paper attributes fund-level AUM and value added equally across comanagers.

Interpretation: This attribution rule is important because it makes manager-level value added measurable even when funds are comanaged. The authors also cluster standard errors by manager and comanagement block because comanaged funds mechanically induce correlation across managers.

3.2 Data sources

The data build on Berk and van Binsbergen (2015). The sample covers monthly observations from January 1977 to March 2011. The authors combine:

- CRSP survivorship-bias-free mutual fund data;
- Morningstar Principia;
- Morningstar Direct manager and firm information;
- manually and algorithmically cleaned manager names and fund-family identifiers.

Interpretation: The data construction is a central part of the paper because manager histories are messy. Manager names can be missing, inconsistent, replaced by “team managed,” or recorded differently across databases.

3.3 Sample restrictions

The paper removes observations without identifiers, missing returns, missing AUM, missing expense ratios, or missing holdings information. It excludes bond funds, money market funds, and index funds. AUM is converted to January 1, 2000 dollars. Different share classes of the same fund are aggregated into one fund.

Interpretation: The goal is to focus on active equity managers whose output can be benchmarked against passive alternatives and whose capital assignment is meaningfully controlled by the fund company.

3.4 Summary scale of the data

The final sample contains 601 firms, 10,423 managers, and 5,542 funds. The industry expands dramatically over the sample: the number of firms rises from 39 in 1978 to 523 in 2011, the number of managers rises from 76 to 4,455, and the number of funds rises from 69 to 3,880.

Interpretation: The sample is large enough to observe many internal reallocations. It also has meaningful variation across firms and over time in organizational scale.

3.5 Definitions of promotions and demotions

A manager is **promoted** when the firm assigns the manager additional funds, increasing that manager’s AUM. A manager is **demoted** when the firm removes funds, decreasing that manager’s AUM. If both adding and removing funds occur simultaneously, the paper uses the net change in assets to classify the event.

Interpretation: These definitions are deliberately internal. The goal is not to study generic AUM growth from investor inflows, but firm decisions that change the manager’s responsibilities.

3.6 Benchmark returns

The cost of capital is the return on the next-best passive opportunity. For fund i at time t , the benchmark return is

$$R_{it}^B = \sum_{j=1}^{n(t)} \beta_i^j R_t^j, \quad (1)$$

where R_t^j is the return on Vanguard index fund j and β_i^j is obtained from a linear projection onto orthogonalized Vanguard index funds.

Interpretation: The benchmark is not an arbitrary factor model. It is meant to represent the passive portfolios actually available to investors, including fees and transaction costs.

3.7 Fund-level value added

Let R_{it}^g be fund i 's gross return before fees and expenses, and let q_{it-1} be lagged AUM. Realized fund-level value added is

$$V_{it} = q_{it-1} (R_{it}^g - R_{it}^B). \quad (2)$$

Interpretation: The measure is in dollars, not percentage returns. A manager adds value only if the capital under management earns more than the passive benchmark before fees.

3.8 Firm-level and manager-level value added

Firm-level value added is the sum over the firm's funds:

$$V_{ft} = \sum_{i \in \Omega_{f,t-1}} V_{it}. \quad (3)$$

Manager-level value added is the sum of the manager's assigned shares of fund value added:

$$V_{mt} = \sum_{i \in \Omega_{m,t-1}} \frac{V_{it}}{n_{it-1}}, \quad (4)$$

where n_{it} is the number of managers of fund i . Manager AUM is similarly

$$q_{mt} = \sum_{i \in \Omega_{m,t}} \frac{q_{it}}{n_{it}}. \quad (5)$$

Interpretation: This lets the authors ask whether reallocating assets to a manager changes the manager's dollar value added.

4 Theory: why a firm can add value

4.1 Basic employee production

Employee i creates value by managing capital q_i . The value created is

$$V_i = q_i \psi_i(q_i), \quad (6)$$

where $\psi_i(q_i)$ is the average excess return on capital. The key assumption is decreasing returns to scale:

$$\psi'_i(q) < 0. \quad (7)$$

Interpretation: A skilled manager may have valuable ideas, but those ideas become harder to scale. Giving a manager more assets can lower percentage performance while still increasing dollar value added.

4.2 Coasian benchmark

In a frictionless world without asymmetric information, outside investors can allocate capital to each employee directly. If employee i works alone, investors choose q_i to maximize

$$\max_{q_i} [q_i \psi_i(q_i) - W_i, 0]. \quad (8)$$

The first-order condition is

$$\psi_i(q_i^*) + q_i^* \psi'_i(q_i^*) = 0. \quad (9)$$

The wage is pinned down by the employee's outside option:

$$W_i = q_i^* \psi_i(q_i^*). \quad (10)$$

If employees work inside a firm, the firm chooses the same q_i^* for each employee and pays the same wage. The firm adds no value.

Interpretation: The null hypothesis is not that firms do not exist. It is that, absent frictions and information asymmetry, the firm cannot improve capital allocation relative to outside investors.

4.3 Asymmetric information

The alternative hypothesis is that the firm executive observes employee quality better than outside investors. Outside investors know only average productivity:

$$\psi(q) = E[\psi_i(q)]. \quad (11)$$

If employees work alone, outside investors assign the same amount of capital $\hat{q}$ to everyone:

$$\psi(\hat{q}) + \hat{q}\psi'(\hat{q}) = 0. \quad (12)$$

The common wage is

$$\bar{W} = \hat{q}\psi(\hat{q}). \quad (13)$$

Inside a firm, the executive can condition capital assignments on private information and solves

$$\Delta V = \max_{\{q_i\}} \sum_{i=1}^I q_i \psi_i(q_i) - I\bar{W} \quad \text{s.t.} \quad \sum_{i=1}^I q_i = Q. \quad (14)$$

Allowing the firm also to choose Q yields

$$\Delta V = \max_{\{q_i, Q\}} \sum_{i=1}^I q_i \psi_i(q_i) - I\bar{W} \quad \text{s.t.} \quad \sum_{i=1}^I q_i = Q. \quad (15)$$

At the optimum,

$$\Delta V = \sum_{i=1}^I \tilde{q}_i \psi_i(\tilde{q}_i) - I\bar{W} \geq 0. \quad (16)$$

Interpretation: The firm creates value because uniform outside allocation is feasible but not generally optimal. If the firm knows which employees are underfunded or overfunded, it can reallocate capital to increase total value.

4.4 The mutual fund example and the alpha trap

The paper's simple example assumes

$$\psi(q) = a - bq. \quad (17)$$

Value added is

$$q\psi(q) = q(a - bq). \quad (18)$$

The optimal capital assignment is

$$q^* = \frac{a}{2b}, \quad \psi(q^*) = \frac{a}{2}, \quad V^* = \frac{a^2}{4b}. \quad (19)$$

If outside investors underestimate the manager and assign only

$$q^I = \frac{a}{4b}, \quad (20)$$

then

$$\psi(q^I) = \frac{3a}{4}, \quad V^I = \frac{3a^2}{16b}. \quad (21)$$

A firm promotion from q^I to q^* raises value added but lowers gross alpha from $3a/4$ to $a/2$.

Interpretation: A fall in alpha after a promotion is not evidence that the promotion destroyed value. It is the expected consequence of scaling a skilled manager toward the value-maximizing point.

5 Empirical specifications and identification

5.1 Baseline internal reallocation regression

The baseline empirical model is

$$V_{mt} = \lambda_t + \lambda_m + \lambda_f + \beta 1_{mt}^{reallocated} + \varepsilon_{mt}. \quad (22)$$

Here V_{mt} is manager value added, λ_t are year-month fixed effects, λ_m are manager fixed effects, λ_f are firm fixed effects, and $1_{mt}^{reallocated}$ equals one after the manager is internally promoted or demoted.

Interpretation: Manager fixed effects control for persistent skill. Firm fixed effects control for persistent firm-level differences. Month fixed effects control for aggregate market and industry shocks. The coefficient β asks whether value added changes after an internal reallocation within the same manager.

5.2 Promotion/demotion split

The paper also estimates

$$V_{mt} = \lambda_t + \lambda_m + \lambda_f + \beta_P 1_{mt}^{promotion} + \beta_D 1_{mt}^{demotion} + \varepsilon_{mt}. \quad (23)$$

Interpretation: Promotions and demotions are conceptually different. Promotions allocate more capital to a manager; demotions remove capital that may have been excessive. Both can increase total value added if they move the manager closer to the optimal scale.

5.3 Controls

The main tables add lagged AUM, manager tenure, and comanager controls. The authors also classify promotions/demotions using dollar fees rather than AUM.

Interpretation: These checks address concerns that the estimates mechanically reflect manager scale, career stage, comanagement, or changes in firm compensation rather than real improvements in value added.

5.4 Adjusted-for-firings specification

If a manager leaves the database after a demotion, the authors fill in value added as zero for subsequent observations. This captures the idea that removing all responsibilities can itself be an extreme demotion.

Interpretation: This makes demotions more comparable to promotions and prevents the sample from ignoring managers who disappear because their funds were removed.

5.5 Counterfactual investor-flow approach

A positive coefficient on internal reallocation shows that the reallocation creates value, but some of this value might have been achieved eventually by investors reallocating capital through fund flows. The authors therefore build counterfactuals in which investors, not firms, would have reallocated capital based on different flow rules.

Interpretation: This provides a conservative lower bound on the firm's unique contribution. Even under very favorable assumptions for investors, firm reallocation still accounts for more than 30% of total value added.

5.6 Investor response regression

Firm compensation is

$$\Pi_{ft} = \sum_{i \in \Omega_{ft}} q_{it} \phi_{it}, \quad (24)$$

where ϕ_{it} is the expense ratio charged by fund i . The paper estimates

$$\Delta \Pi_{ft} = \lambda_f + \lambda_t + \beta 1_{ft}^{reallocate} + \delta tenure_{ft} + \sum_{s=0}^1 \gamma_s (R_{ft-s}^n - R_{ft-s}^B) + \epsilon_{ft}. \quad (25)$$

Interpretation: The coefficient β measures whether the firm's fees rise after it makes a capital reallocation decision, holding returns fixed. A positive coefficient means investors reward the firm for reallocating capital.

5.7 Public-information predictability tests

The paper estimates probit models for whether an event is a firm promotion/demotion or an investor-flow event. It uses realized alpha over the last 1–6 months, 7–18 months, and 19+ months:

$$Pr [1_{mt}^{reallocation\ event} = 1] = \Phi (\beta_0 + \beta_6 \hat{\alpha}_{mt}^6 + \beta_{18} \hat{\alpha}_{mt}^{18} + \beta_T \hat{\alpha}_{mt}^T). \quad (26)$$

Interpretation: If investors could replicate firm decisions with public performance data, past alpha would strongly predict firm reallocations. It does not. It strongly predicts investor flows, but only weakly predicts firm reallocations.

5.8 External reallocation test

The paper repeats the main value-added regressions for external promotions and demotions: cases where managers switch firms and their AUM increases or decreases.

Interpretation: External moves are less likely to use the same internal private information as within-firm reallocations. The absence of significant value-added effects supports the internal-information mechanism.

6 Section-by-section notes

6.1 Section I: Relation to literature

The paper connects several literatures. It relates to theories of the firm, internal capital markets, personnel economics, mutual fund families, delegation, and asset management. The key distinction is that most internal capital market papers study capital reallocation across projects or divisions. This paper studies capital reallocation across employees, with employee output directly observable.

Interpretation: The literature section frames mutual fund firms as a rare setting in which the output of individual employees can be measured and the firm's capital-ownership role is minimal.

6.2 Section II: Theory

Section II formalizes two worlds. In the Coasian benchmark, outside investors observe skill and allocate capital efficiently. Firms add no value. In the asymmetric-information world, investors cannot perfectly distinguish managers; firm executives can. The firm therefore adds value by moving capital from overfunded managers to underfunded managers.

Interpretation: The model does not require firms to own capital. It requires firms to observe more than outsiders and have authority to change capital assignments.

6.3 Section III: Value added of mutual fund firms

Section III defines value added at the fund, manager, and firm levels. This is the measurement core of the paper. Value added is dollars of benchmark-adjusted gross performance, not alpha. It also explains why a firm contribution can be bounded: if investors can already invest in the fund, then the change caused by the firm decision provides a lower-bound estimate of what the firm adds.

Interpretation: The logic is deliberately conservative. If investors invest more in funds belonging to firms because firms monitor and reassign managers, then using investor standalone allocations understates the firm’s true contribution.

6.4 Section IV: Data set

Section IV constructs the mutual fund manager-firm-fund panel. The paper combines multiple datasets and cleans manager identities. The resulting sample is large and spans several decades. It shows an industry with many managers and funds, but substantial concentration in top firms.

Interpretation: This section is essential because the empirical design requires observing managers before and after reallocations and distinguishing internal reallocations from investor flows.

6.5 Section V: Matching capital with skill

Section V estimates the main effect of internal reallocation on value added. The authors avoid an endogenous manager-firm matching problem by looking within managers and within firms around internal decisions. The main results reject the null that capital reallocation adds no value.

Interpretation: This is the paper’s main empirical section. It shows that the firm does not merely coordinate or market funds; it actively increases manager productivity by adjusting AUM responsibilities.

6.6 Section VI: Investor response to capital reallocation

Section VI asks whether firms capture value. If investors understand the signal in capital reallocation, they should allocate more money to the firm after observing reallocations. The paper finds that firm compensation rises because AUM rises, not because expense ratios rise.

Interpretation: Investors appear to treat reallocation as information about the firm’s managerial assignment quality. This is why firms can appropriate some rents from better capital-labor matching.

6.7 Section VII: Source of firm skill

Section VII tests whether the firm is using private information. External reallocations do not produce significant value-added effects. Public performance predicts investor flows much more than firm reallocations. Publicly available variables add little predictive power for firm events, except tenure.

Interpretation: These tests support the interpretation that the firm uses information outsiders do not have. The tenure result is also intuitive: the firm’s informational advantage is stronger when managers are younger and less well known publicly.

6.8 Section VIII: Gross alpha

Section VIII repeats the main analysis using gross alpha instead of value added. Promotions reduce gross alpha and demotions increase it. This is consistent with decreasing returns to scale and with the theory’s warning that alpha is not the right measure of firm value creation.

Interpretation: Gross alpha can move in the opposite direction from value added. This is one of the paper's most important measurement lessons.

6.9 Section IX: Conclusion

The conclusion argues that firms exist partly because they allocate capital to labor efficiently. The mutual fund industry provides a clean test because firms do not own most of the capital and because employee output is directly observable.

Interpretation: The broader lesson is that organizational value can come from information and assignment, not only ownership of assets.

7 Tables and figures

7.1 Figure 1 and Table I: theory intuition and sample scale

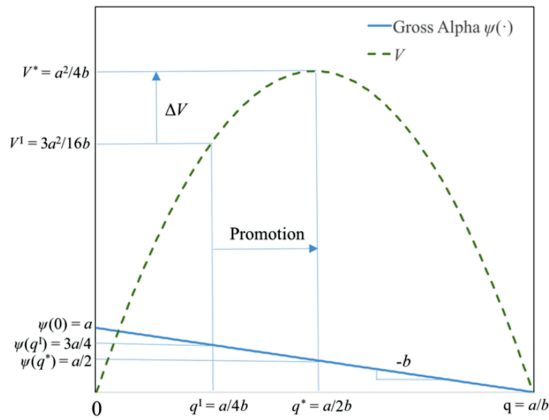
Read: Figure 1:

- Gross alpha $\psi(q) = a - bq$ declines with AUM.
- Value added $q\psi(q)$ rises and then falls, with an interior optimum at $q^* = a/(2b)$.
- A promotion from underfunded capital $q^I = a/(4b)$ to q^* increases total value added but lowers gross alpha.

Read: Table I:

- The sample grows from 39 firms, 76 managers, and 69 funds in 1978 to 523 firms, 4,455 managers, and 3,880 funds in 2011.
- Firm complexity increases: managers per firm rise from 1.95 to 11.6, and funds per firm rise from 1.77 to 7.42.
- Average firm AUM grows from \$0.69 billion in 1978 to \$7.42 billion in 2011.

Economic message: The figure explains why value added is the correct object. The table shows that firms are large enough and complex enough for internal capital allocation to matter.



Value added and gross alpha. The blue solid line plots gross alpha as :

(a) Value added and gross alpha.

Table I
Characteristics of Mutual Fund Firms and Managers

This table reports characteristics of mutual fund firms and managers for selected years in our data set. When a fund is comanaged by N managers, we attribute $\frac{1}{N}$ of the fund's AUM to each of its managers.

Year	Total Number of			Average Number of				Average AUM per		
	Firms	Managers	Funds	Managers per		Funds per		(Y2000 \$ Billion)		
				Firm	Fund	Firm	Manager	Firm	Manager	Fund
1978	39	76	69	1.95	1.23	1.77	1.12	0.69	0.36	0.39
1981	49	108	99	2.20	1.29	2.02	1.19	0.76	0.34	0.38
1984	69	174	157	2.54	1.37	2.28	1.24	0.93	0.37	0.41
1987	116	332	292	2.89	1.44	2.52	1.27	1.26	0.44	0.50
1990	161	552	471	3.48	1.53	2.93	1.30	1.44	0.42	0.49
1993	290	1,083	894	3.80	1.57	3.08	1.30	1.53	0.41	0.50
1996	356	1,789	1,395	5.23	1.77	3.92	1.38	2.99	0.60	0.76
1999	445	2,739	2,124	6.64	1.95	4.77	1.51	4.93	0.80	1.03
2002	501	3,436	2,672	7.83	2.12	5.33	1.65	4.74	0.69	0.89
2005	486	3,651	2,883	9.17	2.32	5.93	1.83	6.27	0.84	1.06
2008	510	4,388	3,621	11.6	2.77	7.10	2.28	8.76	1.02	1.23
2011	523	4,455	3,880	11.6	2.84	7.42	2.47	7.42	0.87	1.00

(b) Characteristics of mutual fund firms and managers.

7.2 Figures 2 and 3: reallocation timing and industry concentration

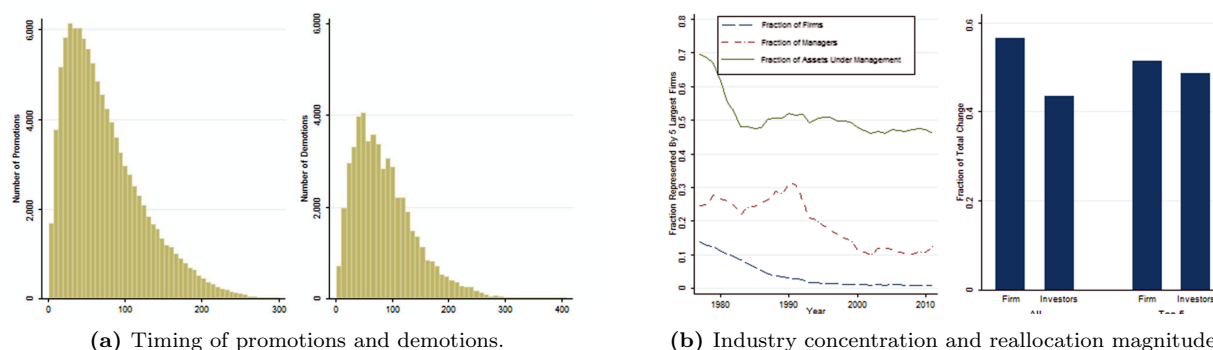
Read: Figure 2:

- Most promotions and demotions occur early in a manager's career.
- Promotions are especially common during the first few years of manager tenure.
- Demotions also cluster early but have a somewhat different distribution.

Read: Figure 3:

- The top five firms account for a small share of firms but a large share of AUM.
- The five largest firms managed 46% of industry assets in January 2011 while employing 13% of managers.
- The bar chart shows that firm capital reallocations are comparable in magnitude to investor flows, especially among top firms.

Economic message: Firm decisions are not rare or trivial. Capital reallocation is an economically important organizational activity in the mutual fund industry.



7.3 Tables II and III: main value-added evidence

Read: Table II:

- The overall capital reallocation coefficient is 0.681, statistically significant at the 99% level.
- Promotions have large positive coefficients: 0.756 in the simple split and about 0.879–0.880 with AUM and tenure controls.
- Demotions are positive but less precisely estimated in the baseline table.
- The results are robust to adding controls and to defining reallocations by fee changes instead of AUM changes.

Read: Table III:

- After adjusting for firings, the capital reallocation coefficient rises to 0.712.
- Promotions and demotions are both positive; demotion coefficients become statistically significant.
- The sample expands to 707,926 observations under the adjusted construction.

Economic message: Internal capital reallocation raises dollar value added. The firm appears to improve productivity by adjusting managers' responsibilities.

Table II
Internal Reallocation of Capital

The dependent variable in the table is V_{mt} defined in (17) as the manager's return in excess of the benchmark times that manager's AUM, measured in \$ millions/month. The first column of the table reports the panel regression specification that uses the capital reallocation dummy from (19). The next column reports the specification where we split reallocations into promotions and demotions, that is, the estimates from (20). The following two columns add lagged AUM (q_{mt-1}) and manager tenure (measured in years since entry into the database) as additional regressors. The fifth column includes as an additional regressor a dummy variable that equals one after a comanager is added to a fund that the manager under consideration manages and remains on until the manager under consideration is either promoted or demoted. The final column defines promotions and demotions based on the change in the total dollar fees collected, rather than the change in AUM. Manager, firm, and year-month fixed effects are included in all regression specifications. Standard errors, two-way clustered by manager and comanagement block, are provided in parentheses. * (**) indicates that the estimate is significantly different from zero at the 95% (99%) confidence level.

Capital Reallocation	0.681** (0.197)					
• Promotion	0.756** (0.250)	0.879** (0.258)	0.880** (0.259)	0.868** (0.263)	0.789** (0.261)	
• Demotion	0.513 (0.432)	0.504 (0.408)	0.510 (0.418)	0.507 (0.414)	0.499 (0.412)	
Comanager				0.576** (0.221)		
AUM		-0.436 (0.442)	-0.433 (0.444)	-0.434 (0.448)		
Manager Tenure			-0.083 (0.110)	-0.091 (0.107)		
Fixed Effects						
• Firm	Yes	Yes	Yes	Yes	Yes	Yes
• Manager	Yes	Yes	Yes	Yes	Yes	Yes
• Year-month	Yes	Yes	Yes	Yes	Yes	Yes
Observations	609,932	609,932	609,932	609,932	609,932	609,932

(a) Internal reallocation of capital.

Table III
Internal Reallocation of Capital (Adjusted for Firings)

The dependent variable in the table is V_{mt} , defined in (17) as the manager's return in excess of the benchmark times that manager's AUM, measured in \$ millions/month. If a manager drops off the database when his demotion dummy is equal to one, we fill in a value added of zero for that manager for all subsequent firm observations. The first column of the table reports the panel regression specification that uses the capital reallocation dummy from (19). The next column reports the specification where we split reallocations into promotions and demotions, that is, the estimates from (20). The final column defines promotions and demotions based on the change in the total dollar fees collected, rather than just the change in AUM. Manager, firm, and year-month fixed effects are included in all regression specifications. Standard errors, two-way clustered by manager and comanagement block, are provided in parentheses. * (**) indicates that the estimate is significantly different from zero at the 95% (99%) confidence level.

Capital Reallocation	0.712** (0.198)		
• Promotion		0.703** (0.225)	0.695** (0.228)
• Demotion		0.720* (0.335)	0.731* (0.337)
Fixed Effects			
• Firm	Yes	Yes	Yes
• Manager	Yes	Yes	Yes
• Year-month	Yes	Yes	Yes
Observations	707,926	707,926	707,926

(b) Internal reallocation adjusted for firings.

7.4 Figures 4–6: lower-bound firm contribution under counterfactual investor flows

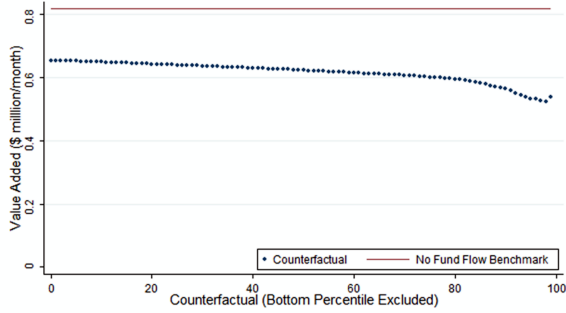
Read: Figures 4 and 5:

- Figure 4 shows value added under a range of counterfactuals that assume investors would have reallocated capital toward successful funds.
- The no-fund-flow benchmark is higher, but even extreme counterfactuals leave substantial value added.
- Figure 5 converts this into a fraction of total value added and shows the firm contributes more than 30% even in conservative cases.

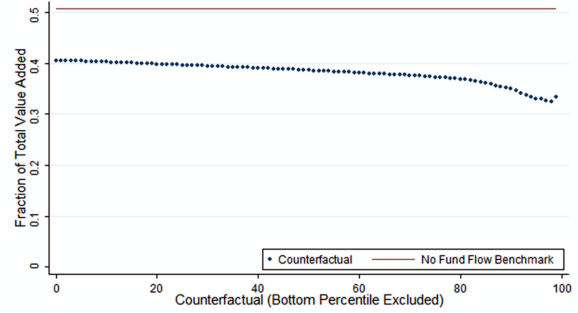
Read: Figure 6:

- The average time for investors to replicate the AUM change from a promotion is measured in years, not months.
- Even the most favorable counterfactual for investors implies that promotions achieve in one month what investor flows would take years to accomplish.

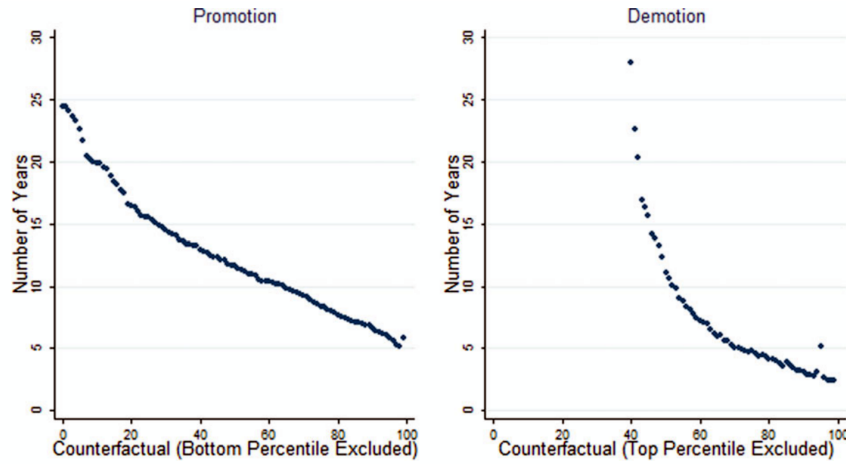
Economic message: The firm is not just doing what investors would have immediately done anyway. It accelerates and improves the allocation of capital to managers.



(a) Value added under counterfactuals.



(b) Firm contribution as a fraction of total value added.



Time needed for investor flows to reproduce firm AUM changes.

7.5 Tables IV and V: investor response and flow-performance relation

Read: Table IV:

- Firm compensation rises by 0.121 after a capital reallocation, statistically significant.
- The effect is not eliminated by controlling for abnormal returns.
- Promotions and demotions both increase firm compensation.
- The increase comes through AUM, not expense ratios: the AUM coefficient is 0.137 while the expense-ratio effect is near zero.

Read: Table V:

- Investor fund flows respond strongly to performance over 3, 6, 12, and 24 months.
- The main performance coefficient β rises with horizon, reaching 1.612 at 24 months.
- The self-employed interaction γ is not robustly positive; investors do not seem more responsive to self-employed managers.

Economic message: Investors use performance to allocate capital, but they also separately respond to firm capital reallocation decisions. The firm captures value because investors allocate additional AUM to the firm.

Table IV

Change in Firm Compensation from Capital Reallocation Decisions

The first two columns of the table report the panel regression specification described in (25), where the change in firm f 's overall compensation in quarter t , $\Delta\Pi_{f,t}$, is regressed on a dummy variable for whether the firm made a capital reallocation decision, a measure of managerial tenure, past abnormal returns, as well as firm and time fixed effects. The next two columns replace the reallocation dummy with a promotion (demotion) dummy that equals one if the firm promotes (demotes) at least one manager during the time period. In the fifth and sixth columns, the dependent variable is replaced with the change in the firm's weighted average expense ratio and overall AUM, respectively. Standard errors, two-way clustered by firm and date, are provided in parentheses. All numbers are reported in \$ millions/month. * (**) indicates that the estimate is significantly different from zero at the 95% (99%) confidence level.

	Change in				Expense Ratio	AUM
	Compensation					
Capital Reallocation	0.121** (0.038)	0.121** (0.038)			-0.004 (0.003)	0.137** (0.038)
• Promotion			0.107** (0.042)			
• Demotion				0.189** (0.053)		
Firm Tenure	0.008 (0.012)	0.008 (0.012)	0.009 (0.012)	0.008 (0.012)	-0.002** (0.001)	0.011 (0.012)
Abnormal Return		1.920** (0.288)	1.900** (0.324)	1.918** (0.282)	0.031 (0.019)	1.910** (0.326)
Lagged Abnormal Return		0.610* (0.295)	0.641* (0.278)	0.606* (0.281)	0.043* (0.019)	0.656* (0.279)
Fixed Effects						
• Firm	Yes	Yes	Yes	Yes	Yes	Yes
• Year-month	Yes	Yes	Yes	Yes	Yes	Yes
Observations	51,807	51,807	51,807	51,807	51,807	51,807

(a) Change in firm compensation from capital reallocation decisions.

Table V

Sensitivity of Fund Flow to Performance

This table reports the coefficient estimates of (27). The dependent variable is the change in the manager's AUM that results from capital flows from investors. Each column in the table reports the results of regressing this measure on the manager's realized return in excess of the benchmark (with coefficient β) as well as an interaction term (with coefficient γ) for whether the manager is self-employed over the past 1, 3, 6, 12, and 24 months. Manager tenure (length of time the manager has been in the database) and year-month fixed effects are included in all the regression specifications. Adjusted R^2 's are also reported. Standard errors, in parentheses, are two-way clustered by manager and comanagement block. * (**) indicates that the estimate is significantly different from zero at the 95% (99%) confidence level.

	1 Month	3 Months	6 Months	12 Months	24 Months
β	0.009 (0.059)	0.345** (0.051)	0.676** (0.055)	1.198** (0.066)	1.612** (0.073)
γ	0.035 (0.064)	-0.101 (0.076)	-0.184 (0.119)	-0.434* (0.205)	-0.522 (0.318)
Manager Tenure	-0.024** (0.004)	-0.025** (0.004)	-0.026** (0.004)	-0.027** (0.004)	-0.027** (0.004)
Year-month	Yes	Yes	Yes	Yes	Yes
Fixed Effect					
Adjusted R^2 (%)	0.08	1.40	2.82	4.75	4.86
Observations	409,873	409,873	409,873	409,873	409,873

(b) Sensitivity of fund flow to performance.

7.6 Tables VI and VII: external reallocations and predictability

Read: Table VI:

- External capital reallocations have no significant value-added effect.
- External promotions and external demotions are both statistically insignificant.
- This contrasts sharply with the internal reallocation results in Tables II and III.

Read: Table VII:

- Past alpha strongly predicts investor flows but explains little of firm internal reallocations.
- The pseudo- R^2 for investor promotions/demotions is 3.70%, but only 0.20% for firm promotions and 0.11% for firm demotions.
- A 1% alpha increase strongly affects investor promotion probabilities but has very small effects on firm promotion probabilities.

Economic message: Firms are not merely mimicking public performance signals. The key value comes from private internal information.

Table VI
External Reallocation of Capital

This table repeats a subset of regression specifications in Table II using external promotions and demotions. The dependent variable in the table is V_{it} , defined in (17) as the manager's return in excess of the benchmark times that manager's (lagged) AUM. An external promotion (demotion) is defined to be a change in the firm the manager works for that is also accompanied by an increase (decrease) in the manager's AUM. The first column of the table reports the panel regression specification in (19) using as the capital reallocation dummy an external capital reallocation decision (that is, an external promotion or demotion, or more simply, a job change). The next column repeats the specification where we split external reallocations into external promotions and external demotions. The final column defines promotions and demotions based on the change in the total dollar fees collected, rather than the change in AUM. Manager and year-month fixed effects are included in all of the regression specifications. Standard errors, two-way clustered by manager and comanagement block, are provided in parentheses. * (**) indicates that the estimate is significantly different from zero at the 95% (99%) confidence level.

Capital Reallocation	0.051 (0.194)		
• Promotion		−0.028 (0.257)	−0.023 (0.256)
• Demotion		0.212 (0.403)	0.198 (0.397)
Fixed Effects			
• Manager	Yes	Yes	Yes
• Year-month	Yes	Yes	Yes
Observations	609,932	609,932	609,932

(a) External reallocation of capital.

Table VII
Predictability of Promotions and Demotions

The dependent variable in Panel A is either a "firm" promotion (demotion) or an "investor" promotion (demotion) dummy variable. As before, a "firm" promotion (demotion) is defined as an increase (decrease) in a manager's AUM that results from a change in the funds he manages. The "investor" promotion (demotion) dummy is equal to one when there is an inflow (outflow) of investor funds and zero otherwise. Panel A of this table reports coefficient estimates and the pseudo- R^2 's for a probit regression of these promotion (or demotion) dummy variables on historical realized alpha (over the past 1 to 6 months, 7 to 18 months, and the remaining history (months 19+)). Standard errors, clustered by comanagement block, are provided in parentheses. * (**) indicates that the estimate is significantly different from zero at the 95% (99%) confidence level. The first column of Panel B of this table reports the fraction of the time each dummy variable is equal to one. It can thus be interpreted as the unconditional probability of a promotion (or demotion). The other columns use the estimates in Panel A and (28) to compute the effect on the unconditional probability of a 0.01 (1%) increase in each regressor while keeping the other regressors fixed.

Panel A: β Estimates				
	pseudo- R^2	1–6 Months	7–18 Months	19+ Months
Promotion				
• Firm (Internal)	0.20%	0.349 (1.035)	6.566** (1.149)	5.875** (1.473)
• Investor	3.70%	22.42** (1.981)	30.97** (1.191)	10.77** (1.316)
Demotion				
• Firm (Internal)	0.11%	−3.418** (1.252)	−3.962 (2.359)	−0.433 (2.203)
• Investor	3.70%	−22.42** (1.981)	−30.97** (1.191)	−10.77** (1.316)
Panel B: Marginal Effects				
	Prob.	1–6 Months	7–18 Months	19+ Months
Promotion				
• Firm (Internal)	0.88%	0.008%	0.167%	0.150%
• Investor	48.1%	8.901%	12.23%	4.291%
Demotion				
• Firm (Internal)	0.65%	−0.060%	−0.069%	−0.008%
• Investor	51.9%	−8.901%	−12.23%	−4.291%

(b) Predictability of promotions and demotions.

7.7 Tables VIII and IX: public information and gross alpha

Read: Table VIII:

- Adding public variables such as fund flows, turnover, expense ratios, and tenure still leaves firm reallocations hard to predict.
- Fund flows are not significant predictors, indicating that investors' prior flows do not reveal the firm event.
- Manager tenure is negative and significant: reallocations are more likely for less seasoned managers.

Read: Table IX:

- Promotions reduce gross alpha by about 0.031–0.036 percentage points per month.
- Demotions increase gross alpha by about 0.031–0.033 percentage points per month.
- These effects are consistent with decreasing returns to scale.

Economic message: Public information has little power to predict firm reallocations, while gross alpha moves in the direction expected under optimal scale adjustment. The correct conclusion is not that promotions destroy skill, but that they scale managers toward more value creation.

Table VIII
Predicting Promotions and Demotions from Publicly Available Information

The table implements (28), a probit regression of a promotion (demotion) dummy on the following publicly observable variables: historical realized alpha (over the last 6 months, the prior 7 to 18 months, and the remaining history (months 19+), fund flow in/out of the manager's funds over the last 6 months (1–6 M), the prior 7 to 18 months (7–18 M), and the total flow before that point (19+ M), manager turnover, expense ratio, and tenure (the length of time (in months) since the manager first entered our sample). Manager turnover and expense ratio are computed as the AUM-weighted average of turnover and expense ratio of all funds under the manager's management. Standard errors, clustered by comanagement block are provided in parentheses. * (**) indicates that the estimate is significantly different from zero at the 95% (99%) confidence level.

	Promotion		Demotion	
Realized Alpha (1–6 M)	0.346 (1.032)	0.344 (1.033)	−3.333** (1.333)	−3.332** (1.329)
Realized Alpha (7–18 M)	6.446** (1.236)	4.333** (1.241)	−3.879 (2.498)	−3.879 (2.493)
Realized Alpha (19+ M)	5.698** (1.521)	5.774** (1.528)	−0.533 (2.006)	−0.528 (2.005)
Fund Flow (1–6 M)	0.282 (1.484)	0.287 (1.486)	0.297 (1.371)	0.299 (1.374)
Fund Flow (7–18 M)	−0.556 (1.461)	−0.241 (1.469)	0.241 (1.228)	0.237 (1.222)
Fund Flow (19+ M)	0.688 (1.312)	0.684 (1.322)	0.359 (1.455)	0.361 (1.458)
Turnover	−0.003 (0.455)	−0.003 (0.456)	0.058 (0.509)	0.058 (0.513)
Expense Ratio	0.002 (0.008)	0.002 (0.008)	0.001 (0.008)	0.001 (0.009)
Manager Tenure		−0.021* (0.010)		−0.040** (0.010)
Pseudo- R^2	0.26%	0.49%	0.16%	0.64%
Observations	609,932	609,932	609,932	609,932

(a) Public-information prediction of firm reallocations.

Table IX
Gross Alpha and Capital Reallocation

This table repeats the analysis reported in Table II with return outperformance, $R_{it}^R - R_{it}^B$, as defined by (21) and (23), replacing value added as the dependent variable. The first column of the table reports the specification where reallocations are split into promotions and demotions, that is, the estimates from (20) with return outperformance instead of value added on the left-hand side. The following two columns add lagged AUM (q_{mt-1}) and manager tenure (measured in years since entry into the database) as additional regressors. Standard errors, two-way clustered by manager and comanagement block, are provided in parentheses. All numbers are in %/month. * (**) indicates that the estimate is significantly different from zero at the 95% (99%) confidence level.

Capital Reallocation			
• Promotion	−0.036** (0.013)	−0.032* (0.013)	−0.031* (0.013)
• Demotion	0.031* (0.016)	0.032* (0.016)	0.033* (0.016)
AUM		−0.017** (0.003)	−0.016** (0.003)
Manager Tenure			−0.012* (0.006)
Fixed Effects			
• Firm	Yes	Yes	Yes
• Manager	Yes	Yes	Yes
• Year-month	Yes	Yes	Yes
Observations	609,932	609,932	609,932

(b) Gross alpha and capital reallocation.

7.8 Appendix Table A.1: benchmark funds

Read:

- The appendix lists Vanguard index funds used to construct passive benchmarks.
- These benchmarks span major equity styles, sizes, international exposures, and balanced strategies.
- The benchmark set is central to converting gross mutual fund performance into value added relative to passive alternatives.

Economic message: The measurement of value added depends on a credible outside option. The Vanguard benchmark system makes the comparison economically interpretable.

The listed ticker is for the Investor class shares that we use until Vanguard introduced an Admiral class for the fund, and thereafter we use the return on the Admiral class shares (Admiral class shares have lower fees but require a higher minimum investment).

Fund Name	Ticker	Asset Class	Inception Date
S&P 500 Index	VFINX	Large-Cap Blend	08/31/1976
Extended Market Index	VEXMX	Mid-Cap Blend	12/21/1987
Small-Cap Index	NAESX	Small-Cap Blend	01/01/1990*

(Continued)

Table A.1—Continued

Fund Name	Ticker	Asset Class	Inception Date
European Stock Index	VEURX	International	06/18/1990
Pacific Stock Index	VPACX	International	06/18/1990
Value Index	VVIAX	Large-Cap Value	11/02/1992
Balanced Index	VBINX	Balanced	11/02/1992
Emerging Markets Stock Index	VEIEX	International	05/04/1994
Mid-Cap Index	VIMSX	Mid-Cap Blend	05/21/1998
Small-Cap Growth Index	VISGX	Small-Cap Growth	05/21/1998

(a) Benchmark Vanguard index funds, part 1.

(b) Benchmark Vanguard index funds, part 2.

8 Identification: what makes the estimates credible

8.1 Why the design avoids the main manager-firm matching problem

A simple firm fixed-effect regression would be hard to interpret because managers choose where to work and firms choose whom to hire. If high-skill managers sort into high-quality firms, firm fixed effects might capture

manager sorting rather than firm value creation. The paper therefore focuses on internal reallocations for the same manager.

Interpretation: The identifying variation is within manager and within firm: a manager's value added before and after the firm changes the manager's responsibilities.

8.2 Why internal reallocations are informative

Internal reallocations are decisions made by the firm. Investors cannot directly assign funds to managers inside a fund family. The event therefore reveals the firm's use of authority and information.

Interpretation: The event is not merely investor flow. It is the firm's choice to alter a manager's job scope.

8.3 Why external reallocations are an important placebo

If all AUM increases mechanically produced more value added, then external promotions should also increase value added. They do not. This suggests the internal result is not a mechanical AUM effect.

Interpretation: The difference between internal and external reallocations supports the private-information hypothesis.

8.4 Why gross alpha results strengthen the interpretation

A promotion lowers alpha but raises value added. This is exactly what the theory predicts when an underfunded manager is moved toward optimal scale. A simple alpha-based evaluation would reach the wrong conclusion.

Interpretation: The paper is not claiming promotions raise percentage returns. It is claiming they raise dollar value created.

8.5 Remaining limitations

- The paper cannot observe every internal signal executives use.
- The exact private information channel is inferred rather than directly measured.
- Mutual funds are an unusually clean setting; other industries may have different contracting, output measurement, and capital-ownership frictions.
- The counterfactual investor-flow analysis is a lower bound, not a full structural model of investor learning.
- The sample is active equity mutual funds and does not directly cover hedge funds, private equity, bank lending, or industrial firms.

Interpretation: The evidence is strong for mutual funds, but the general theory should be tested in additional industries where employee output and job scope can be measured.

9 Short summary

- The paper studies why mutual fund firms exist when investors own the capital and can observe fund returns.
- The central answer is that firms privately observe managerial skill and efficiently match capital to labor.

- The theory shows that, absent asymmetric information, firms add no value; with private information, firms add value by assigning different capital levels to different managers.
- Value added is measured in dollars, not alpha: $V_{it} = q_{it-1}(R_{it}^g - R_{it}^B)$.
- The data cover January 1977 to March 2011 and contain 601 firms, 10,423 managers, and 5,542 funds.
- Internal promotions and demotions are defined by firm-driven changes in a manager's AUM.
- Baseline internal reallocation increases value added by about \$681,000 per manager-month.
- Promotions are strongly positive; demotions become positive and significant after adjusting for firings.
- Counterfactual flow exercises imply the firm contributes at least \$474,000 per month, or more than 30% of average total value added.
- Investors respond to firm reallocations by allocating additional AUM to the firm, increasing firm compensation.
- External reallocations do not have significant value-added effects.
- Public performance data explain investor flows but explain little of firm internal reallocations.
- Public information beyond returns also has little predictive power; manager tenure is the main exception.
- Gross alpha falls after promotions and rises after demotions, consistent with decreasing returns to scale.
- The paper's main lesson is that firms can create value by assigning responsibility, even when they do not own physical capital.

10 Conclusion

10.1 Core conclusion

Berk, van Binsbergen, and Liu show that mutual fund firms add value by matching capital with managerial skill. The firm uses private information to decide how much capital each manager should control. These internal decisions increase manager value added and account for at least 30% of the industry's total value added.

10.2 Economic intuition

The intuition is that outside investors observe performance but not the full internal information set of the fund company. The firm can observe much more about a manager's ability, process, trades, and constraints. When a manager is underfunded, the firm can promote the manager by assigning more capital. When a manager is overfunded, the firm can demote the manager by reducing responsibility. Both actions can raise dollar value added.

10.3 Takeaway

The paper broadens the theory of the firm. Firms do not exist only because they own assets, reduce transaction costs, or write incomplete contracts. They can also exist because they are privately informed allocators of responsibility. In mutual funds, the firm's essential function is to match capital to labor.